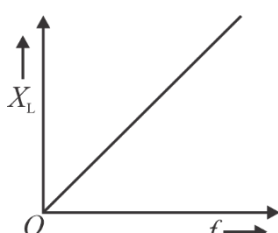


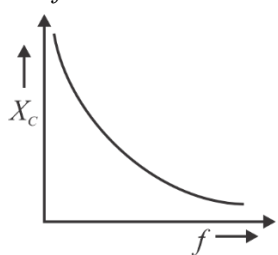
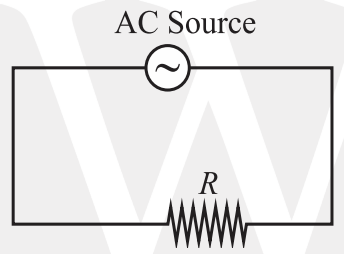
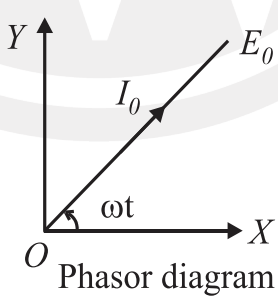


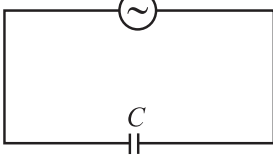
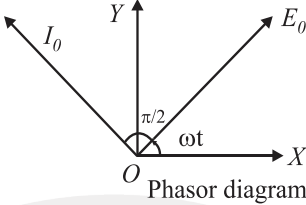
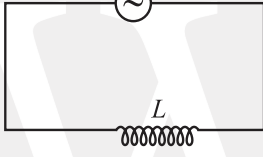
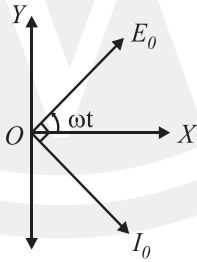
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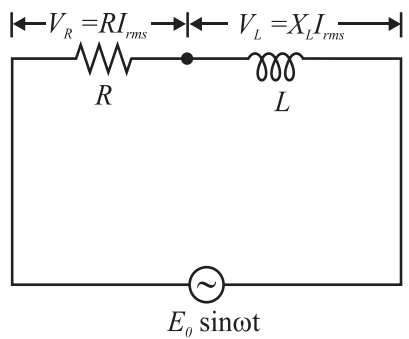
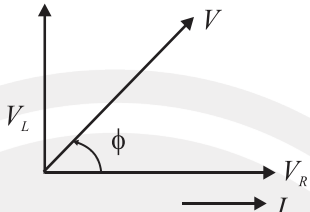
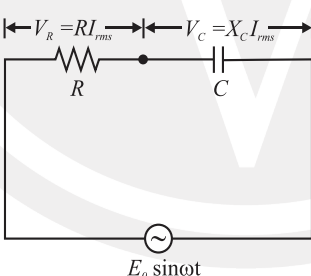
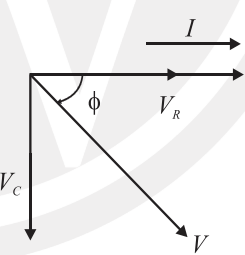
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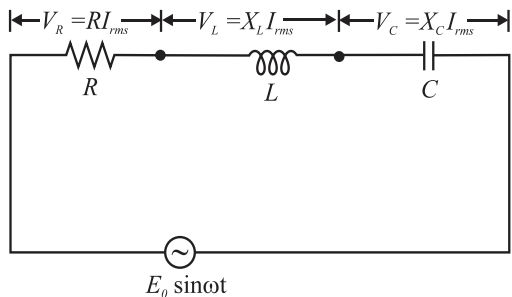
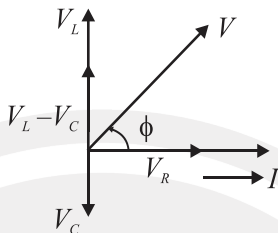
Alternating Current

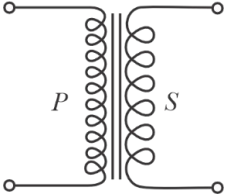
Sr. No.	Concept	Formulas	Other information
1.	Alternating current (A.C)	Instantaneous current: $I = \frac{E}{R} = \frac{E_0}{R} \sin \omega t \quad \text{or,} \quad I = I_0 \sin \omega t$ Where, $I_0 = \frac{E_0}{R} = \text{Peak or maximum value of a.c.}$ Time period of ac current: $T = \frac{2\pi}{\omega}$ Frequency of ac current: $f = \frac{1}{T} = \frac{\omega}{2\pi}$ So, an alternating current can be represented as; $I = I_0 \sin \omega t = I_0 \sin 2\pi f t = I_0 \sin \frac{2\pi}{T} t$	I_0 = the peak value of alternating current. f = frequency of ac current. ω = angular frequency in ac system. t = time. E_0 = the peak value of alternating voltage.
2.	Average and r.m.s value of current and voltage	Average or Mean value of a.c. over half-cycle, $I_{av} = \frac{2}{\pi} I_0 = 0.637 I_0$ Effective or rms or virtual value of a.c., $I_{eff} \text{ or } I_{rms} \text{ or } I_v = \frac{1}{\sqrt{2}} I_0 = 0.707 I_0$ For alternating voltage, we have $E = E_0 \sin \omega t, \quad E_{av} = 0.637 E_0, \quad E_{rms} = \frac{1}{\sqrt{2}} E_0$	I_0 = the peak value of alternating current. E = voltage at any instant E_{av} = average value of alternating voltage. E_{rms} = root mean square value of ac voltage.
3.	Inductive reactance (X_L)	Inductive reactance (X_L) = $L\omega = L 2\pi f$ $= \frac{L 2\pi}{T}$ $X_L \propto f$  For direct current, $X_L = 0$ ($\because f = 0$)	f = frequency of ac current. L = inductance of a inductor. ω = angular frequency in ac system.

Sr. No.	Concept	Formulas	Other information
4.	Capacitive reactance (X_C)	Capacitive reactance, $X_C = \frac{1}{C\omega} = \frac{1}{C2\pi f} = \frac{T}{C2\pi}$  Its unit is ohm. $X_C \propto \frac{1}{f}$ For direct current, $X_C = \infty$ ($\because f = 0$)	f = frequency of ac current. C = capacitance of a capacitor.
5.	Power factor of AC circuit	$V = V_0 \sin \omega t$ $I = I_0 \sin(\omega t + \phi)$ The average power in an AC circuit, $P_{av} = V_{rms} I_{rms} \cos \phi$ $= \frac{V_0}{\sqrt{2}} \frac{I_0}{\sqrt{2}} \cos \phi = \frac{V_0 I_0}{2} \cos \phi$ where, $\cos \phi = \frac{\text{Resistance (R)}}{\text{Impedance (Z)}}$	$\cos \phi$ = power factor of ac circuit. V_{rms} = root mean square – voltage of ac. I_{rms} = root mean square current of ac.
6.	Pure resistive circuit (R only)	  Phasor diagram (i) Alternating emf, $E = E_0 \sin \omega t$ (ii) Alternating current, $I = I_0 \sin \omega t$ (iii) Alternating emf and alternating current both are in phase. (iv) Power factor, $\cos \phi = 1$ Average power (P) = $V_{rms} \cdot I_{rms}$	I_0 = the peak value of alternating current. t = time of measurement. R = resistance of the circuit.

Sr. No.	Concept	Formulas	Other information
7.	Pure capacitive circuit (C only)	AC Source  Phasor diagram  (i) Alternating emf, $E = E_0 \sin \omega t$ (ii) Alternating current, $I = I_0 \sin(\omega t + \frac{\pi}{2})$ (iii) Alternating current leads the alternating emf by $\frac{\pi}{2}$. (iv) Power factor, $\cos \phi = \cos 90^\circ = 0$ Average power, $P_{av} = 0$	C = capacitance of the capacitor. E_0 = the peak value of alternating emf.
8.	Pure inductive circuit (L only)	AC Source  Phasor diagram  (i) Alternating emf, $E = E_0 \sin \omega t$ (ii) Alternating current, $I = I_0 \sin(\omega t - \frac{\pi}{2})$ (iii) Alternating current lags behind alternating emf by $\frac{\pi}{2}$. (iv) Power factor, $\cos \phi = \cos 90^\circ = 0$ Average power, $P_{av} = 0$	ω = angular frequency in ac system. I = instantaneous current at any instant.

Sr. No.	Concept	Formulas	Other information
9.	A.C. through a series <i>LR</i> circuit	  Instantaneous voltage, $E = E_0 \sin \omega t$ Instantaneous current, $I = I_0 \sin(\omega t - \phi)$ Peak current, $I_0 = \frac{E_0}{\sqrt{R^2 + X_L^2}} = \frac{E_0}{Z}$ Impedance, $Z = \sqrt{R^2 + X_L^2} = \sqrt{R^2 + \omega^2 L^2}$ Phase angle, $\phi = \tan^{-1} \frac{X_L}{R}$ or $\phi = \cos^{-1} \frac{R}{Z}$ Resultant voltage $V = \sqrt{V_L^2 + V_R^2}$	L = inductance of the inductor. Z = total impedance of circuit. R = resistance present in the circuit. $I_{rms} = \frac{E_{rms}}{Z}$
10.	A.C. through a series <i>CR</i> circuit	  Instantaneous voltage, $E = E_0 \sin \omega t$ Instantaneous current, $I = I_0 \sin(\omega t + \phi)$ Peak current, $I_0 = \frac{E_0}{\sqrt{R^2 + X_C^2}} = \frac{E_0}{Z}$ Impedance, $Z = \sqrt{R^2 + X_C^2} = \sqrt{R^2 + \omega^2 L^2}$ Phase angle, $\phi = \tan^{-1} \frac{X_C}{R}$ or $\phi = \cos^{-1} \frac{R}{Z}$ Resultant voltage $V = \sqrt{V_R^2 + V_C^2}$	X_C = capacitive reactance. ω = angular frequency. E_0 = peak value of the voltage. $I_{rms} = \frac{E_{rms}}{Z}$

Sr. No.	Concept	Formulas	Other information
11.	A.C. through a series LCR circuit	  $I = I_0 \sin(\omega t - \phi) = \frac{E_0}{Z} \sin(\omega t - \phi)$ Its impedance; $Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$ The phase angle ϕ between voltage and current is given by $\tan \phi = \frac{X_L - X_C}{R} \text{ and } \cos \phi = \frac{R}{Z}$ Resultant voltage $V = \sqrt{V_R^2 + (V_L - V_C)^2}$	Z = impedance of the circuit L = inductance of the inductor C = capacitance of the capacitor. $I_{rms} = \frac{E_{rms}}{Z}$
12.	Resonance condition of the LCR-circuit	$X_L = X_C$, the impedance of LCR circuit becomes $Z = \sqrt{R^2 + (X_L - X_C)^2} = R$ Resonant frequency: $f_r = \frac{1}{2\pi\sqrt{LC}}$	f_r = resonant frequency. R = resistance of the circuit.
13.	Q-Factor of LCR circuit	Q factor = $\frac{\text{Resonant frequency}}{\text{Band width}}$ $= \frac{\omega_r}{\omega_2 - \omega_1} \text{ or }$ $Q = \frac{V_L \text{ or } V_C}{V} \text{ or } Q = \frac{X_L}{R} \text{ or } \frac{X_C}{R}$ $\text{or } Q = \frac{\omega_r L}{R} \text{ or } Q = \frac{1}{C\omega_r R}$ $\text{or } Q = \frac{1}{R} \sqrt{\frac{L}{C}}$	Q = quality factor of the circuit. ω_2 and ω_1 = the frequencies at which the current falls to $\frac{1}{\sqrt{2}}$ times of its resonant value.

Sr. No.	Concept	Formulas	Other information
14.	Transformer	For an ideal transformer, $\frac{E_2}{E_1} = \frac{I_1}{I_2} = \frac{N_2}{N_1} = k$  (i) For a step-up transformer, $k > 1$ or $N_2 > N_1$ $\therefore E_2 > E_1$ and $I_2 < I_1$. (ii) For a step-down transformer, $k < 1$ or $N_2 < N_1$ $\therefore E_2 < E_1$ and $I_2 > I_1$.	E_2, E_1 = voltage in the secondary and primary coil. N_2, N_1 = number of turns of the secondary coil and primary coil. k = turn ratio of transformer. I_2, I_1 = current in the secondary and primary coil